EE 354/CE 361/MATH 310 – Introduction to Probability and Statistics

Fall 2025

EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

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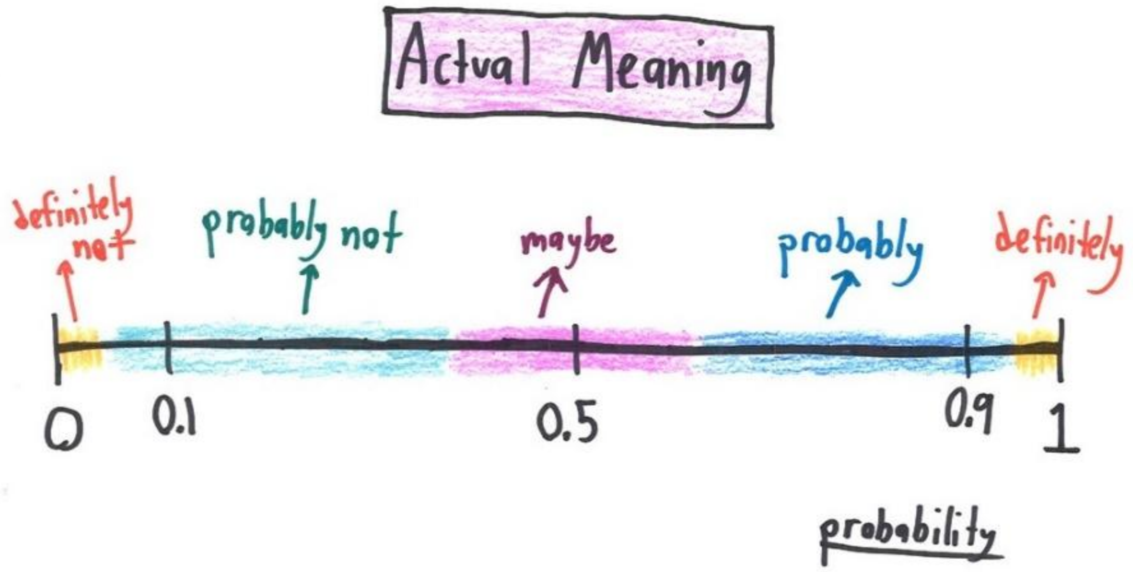
Week 1 - Lecture No 02 – 25th August 2025



Announcements!

1. Lecture No 01 posted
2. Supplemental Reading on set theory posted (for Unit No 1) – please go through it
3. Quiz no 01 – next week on September 01, 2025 from Unit No 01

Agenda for today

1. Unit 1: Probability models and axioms

Unit 1: Probability models and axioms Lecture outline

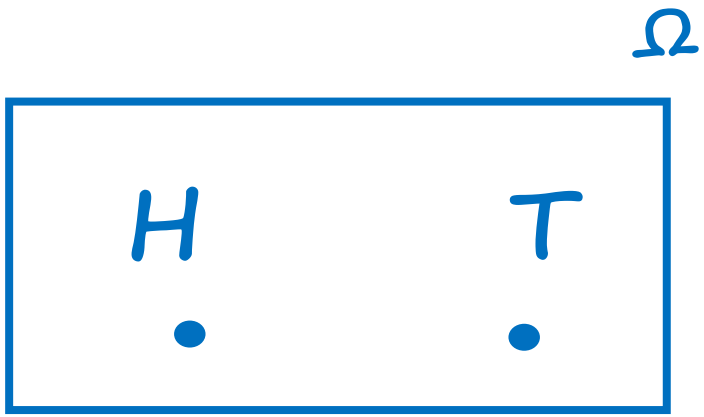
- Sample space
- Probability laws
  - Axioms
  - Some properties
- Examples
  - Discrete
  - Continuous
- Interpretations of probabilities

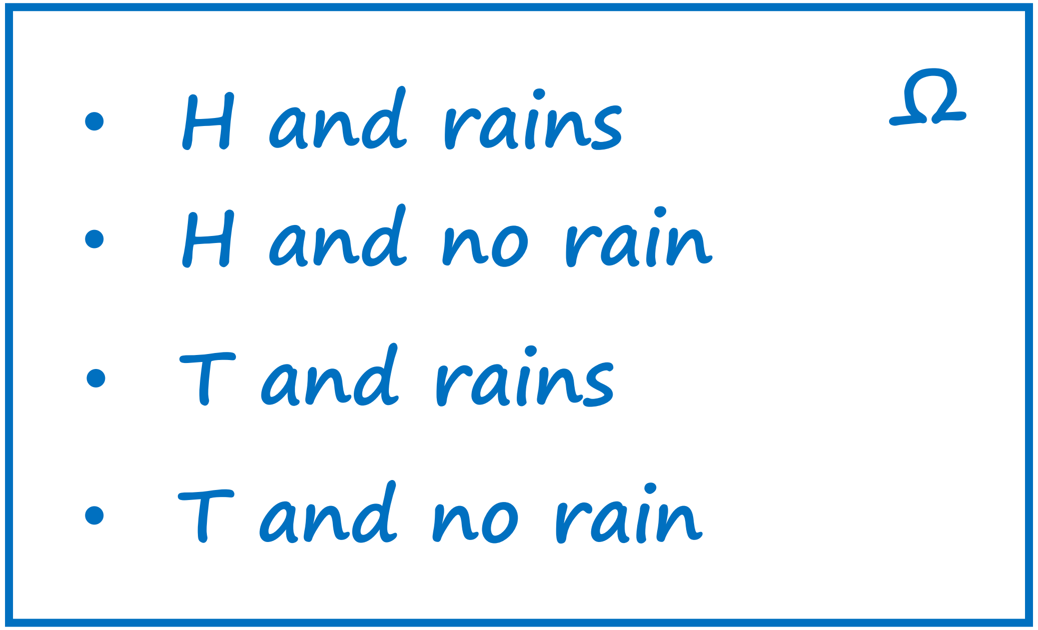
Sample space

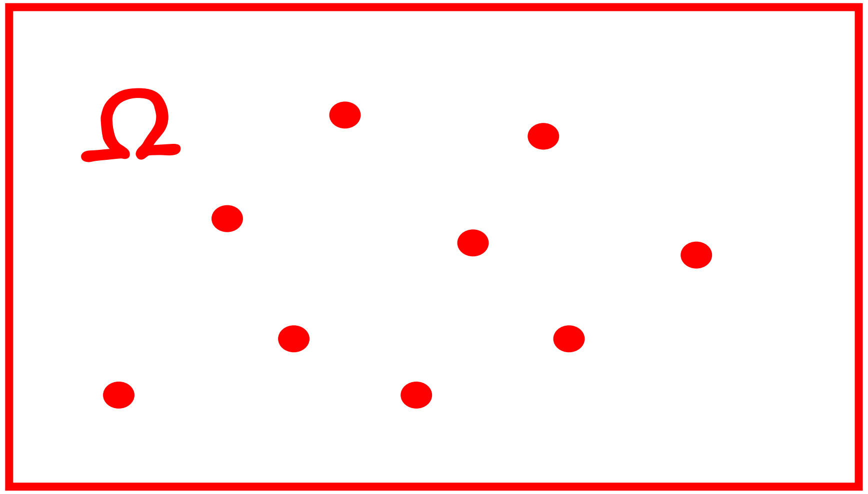
- Two steps
  - Describe possible outcomes
  - Describe beliefs about likelihood of outcomes

Sample space

- List (set) of possible outcomes, Ω
- List must be:
  - Mutually exclusive
  - Collectively exhaustive
  - At the "right granularity"

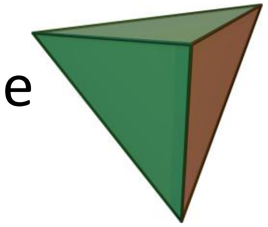


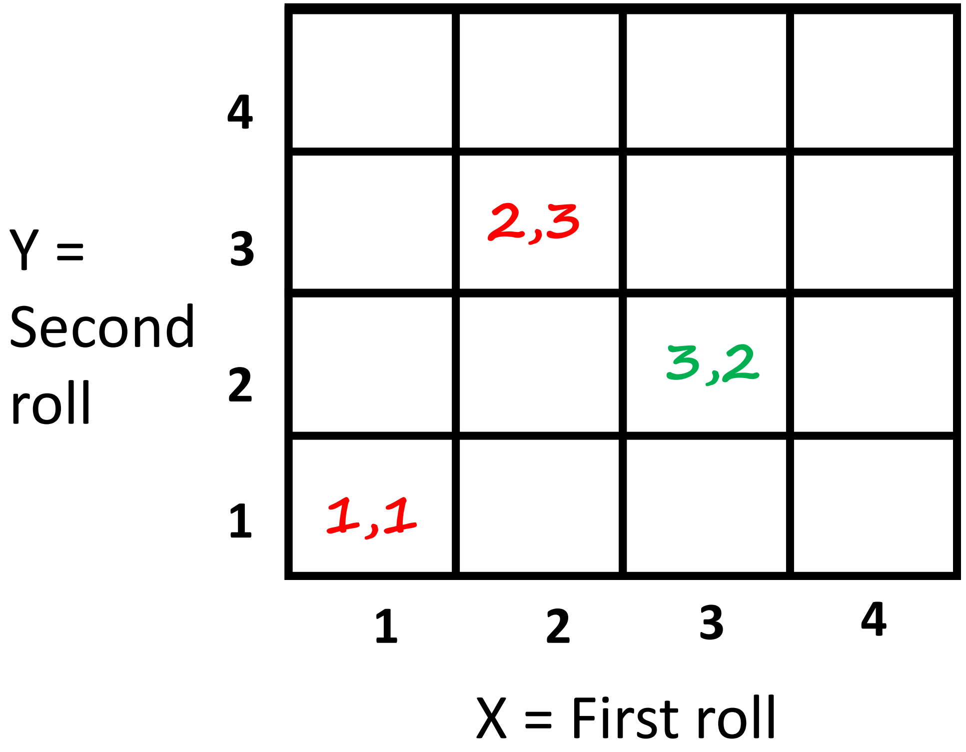




Sample space: discrete/finite example

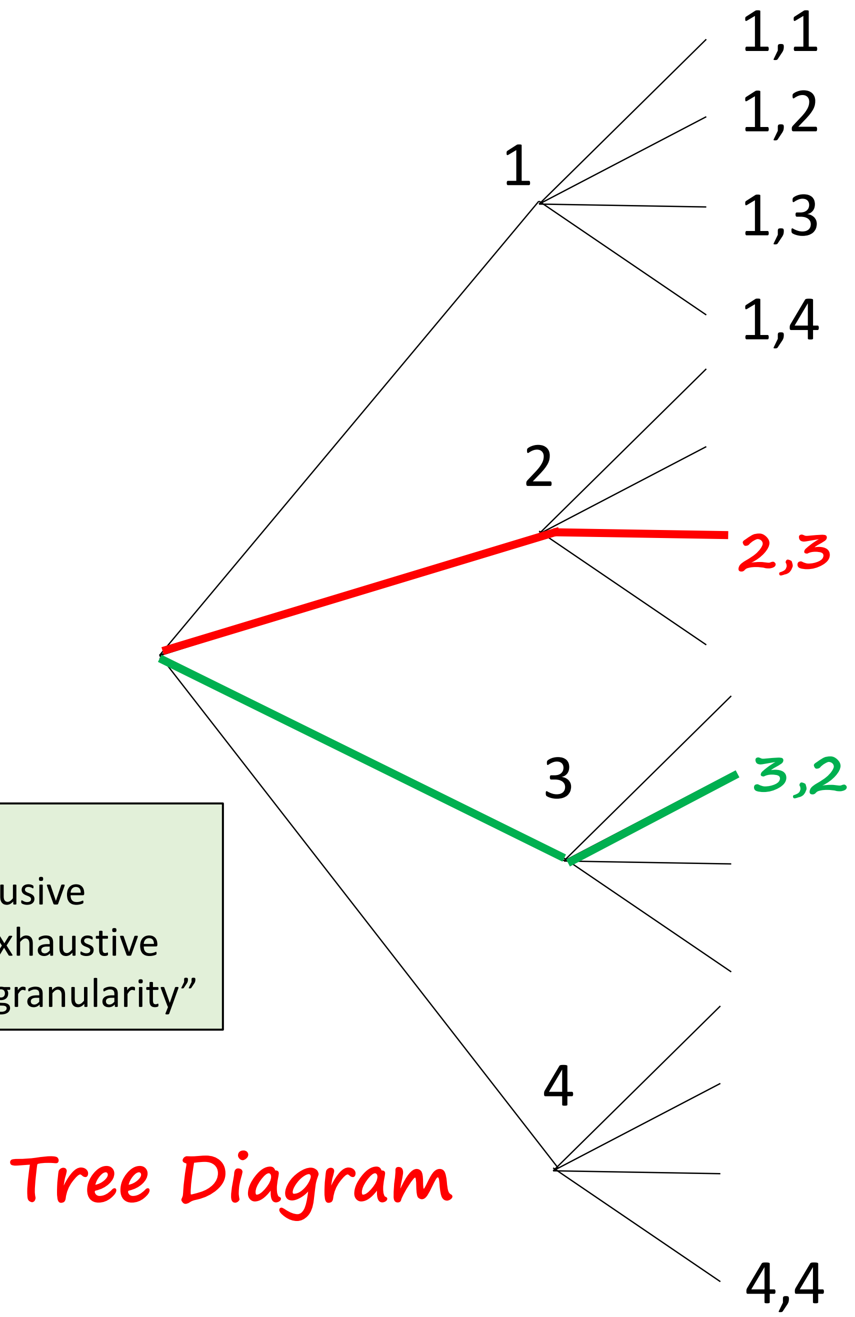
- Two rolls of a tetrahedral die





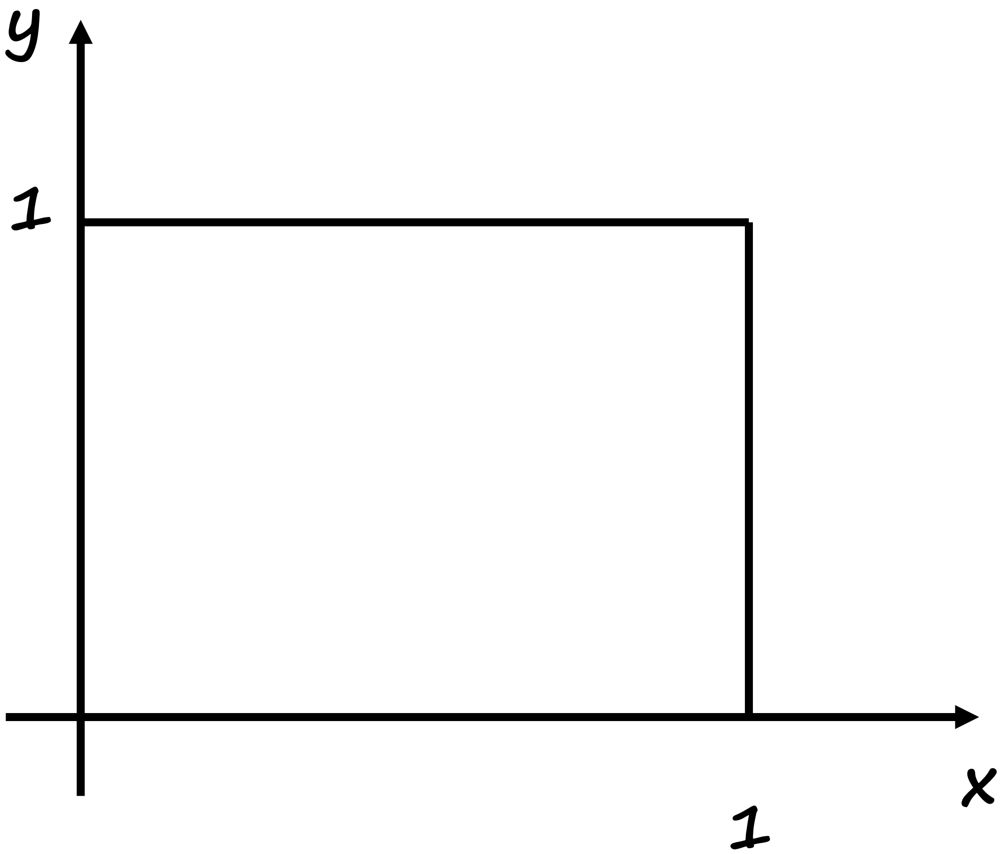
- List must be:
  - Mutually exclusive
  - Collectively exhaustive
  - At the "right granularity"

Sequential description



Sample space: continuous example

- (x,y) such that $0 \le x,y \le 1$

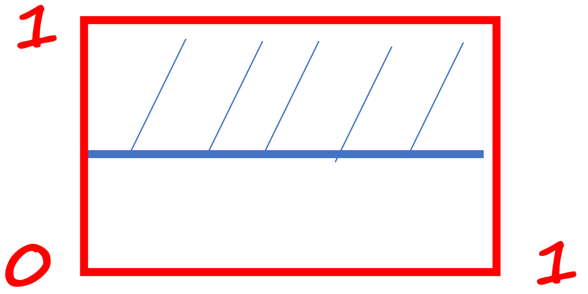


Probabilistic Model

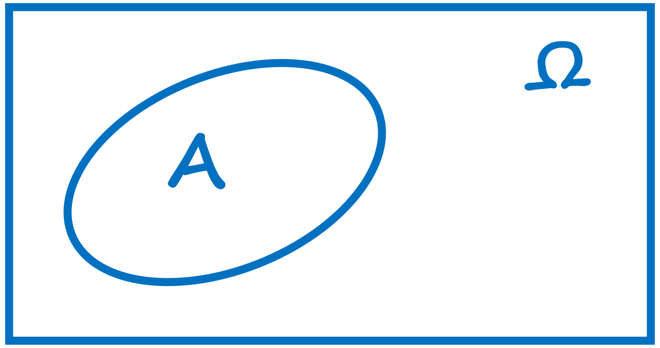
- Two steps
  - ➢ Describe possible outcomes – Sample Space
  - ➢ **Describe beliefs about likelihood of outcomes -**

Probability axioms

- Event: a subset of the sample space
  - Probability is assigned to events



→ P (A)

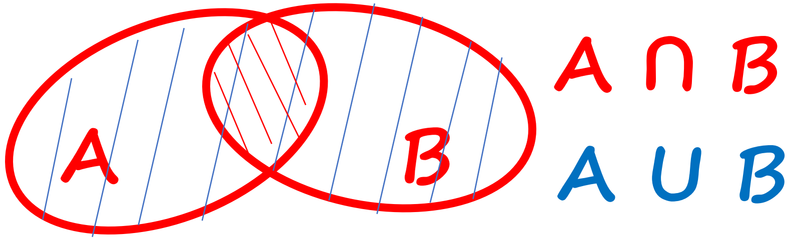


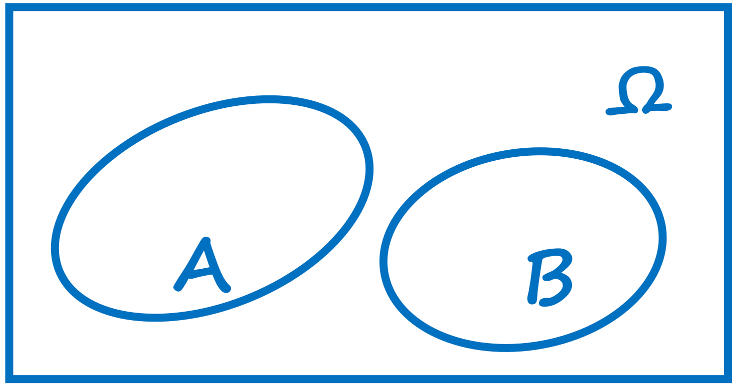
- Axioms:
  - Nonnegativity: $\mathbf{P}(A) \geq 0$
  - Normalization: $\mathbf{P}(\Omega) = 1$
  - (Finite) additivity: (to be strengthened later)

    If $A \cap B = \emptyset$, then $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$

    ↑

    empty set





Some simple consequences of the axioms

Axioms

$P(A) \geq 0$

$P(\Omega) = 1$

For disjoint events:

$P(A \cup B) = P(A) + P(B)$

Consequences

$P(A) \leq 1$

$P(\emptyset) = 0$

$P(A) + P(A^c) = 1$

$P(A \cup B \cup C) = P(A) + P(B) + P(C)$

and similarly for *k* disjoint events

$$P(\{s_1, s_2, \dots, s_k\}) = P(\{s_1\}) + \dots + P(\{s_k\})$$
$$= P(s_1) + \dots + P(s_k)$$

Some simple consequences of the axioms

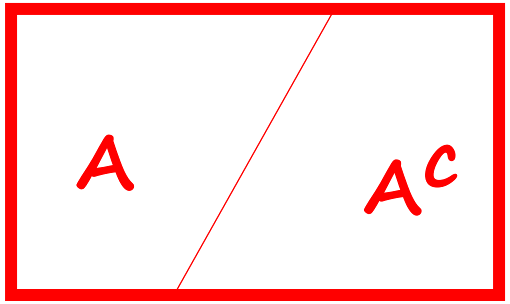
Axioms

a) $P(A) \geq 0$

b) $P(\Omega) = 1$

For disjoint events:

c) $P(A \cup B) = P(A) + P(B)$



$A \cup A^c = \Omega$

$A \cap A^c = \emptyset$

$\Rightarrow 1 \overset{(b)}{=} P(\Omega) = P(A \cup A^c)$

$\Rightarrow \overset{(c)}{=} P(A) + P(A^c)$

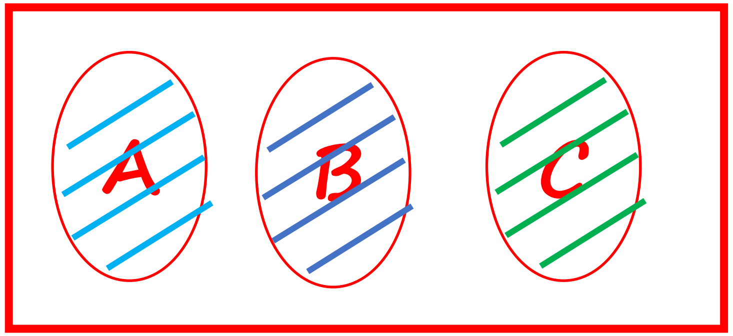
$\Rightarrow P(A) = 1 - P(A^c) \overset{(a)}{\leq} 1$

$\Rightarrow 1 = P(\Omega) + P(\Omega^c)$

$\Rightarrow 1 = 1 + P(\emptyset) \Rightarrow P(\emptyset) = 0$

Some simple consequences of the axioms

- A, B, C disjoint: P (A U B U C) = P (A) + P (B) + P (C)

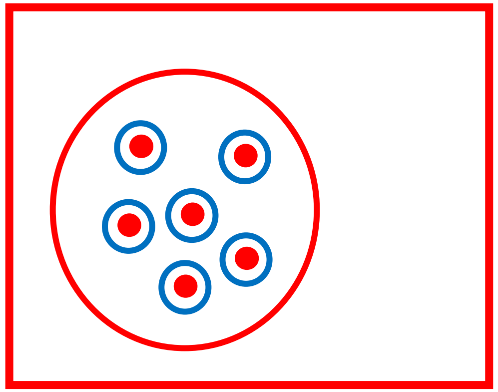


$P(A \cup B \cup C) = P((A \cup B) \cup C) = P(A \cup B) + P(C)$

$\Rightarrow \quad = P(A) + P(B) + P(C)$

$\Rightarrow$ If $A_1, \ldots A_k$ disjoint $\Rightarrow P\left(A_1 \cup \ldots \cup A_k\right) = \sum_{i=1}^{k} P(A_i)$

- P $(\{s_1, s_2, \ldots, s_k\})$ = $P(\{s_1\} \cup \{s_2\} \cup \ldots \cup \{s_k\})$

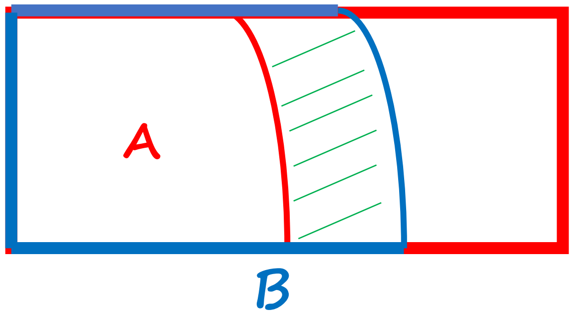


$= P(\{s_1\} + \{s_2\} + \ldots + \{s_k\})$

$= P(s_1) + P(s_2) + \ldots + P(s_k)$

More consequences of the axioms

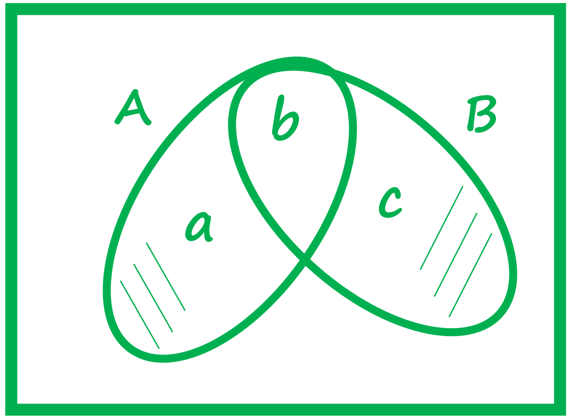
- If $A \subset B$, then $\mathrm{P}(A) \leq \mathrm{P}(B)$



$\Rightarrow B = A \cup (B \cap A^c)$

$P(B) = P(A) + \underline{P(B \cap A^c)} \geq P(A)$

- $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \overline{\mathrm{P}(A \cap B)}^{\,\geq 0}$



$a = P(A \cap B^c) \qquad b = P(A \cap B) \qquad c = P(B \cap A^c)$

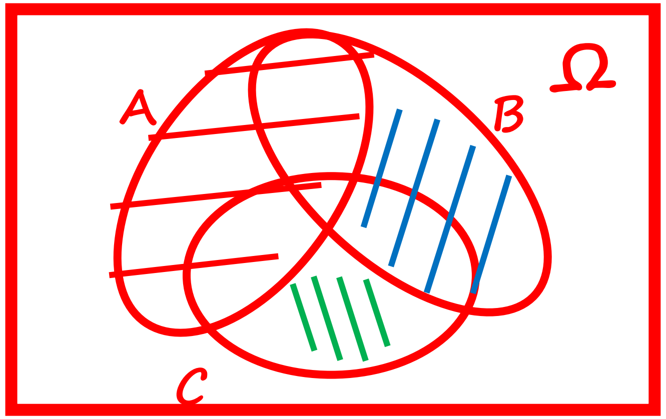
$P(A \cup B) = a + b + c$

$$P(A) + P(B) - P(A \cap B) = (a + b) + (b + c) - b$$
$$= a + b + c$$

- $\mathrm{P}(A \cup B) \leq \mathrm{P}(A) + \mathrm{P}(B)$ union bound

More consequences of the axioms

- $P(A \cup B \cup C) = P(A) + P(\underline{A^C \cap B}) + P(\underline{A^C \cap B^C \cap C})$

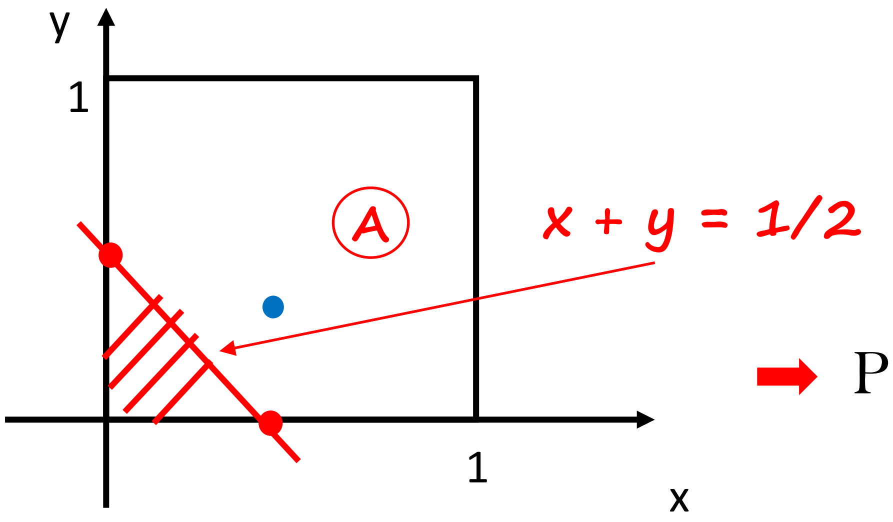


$\Rightarrow P(A \cup B \cup C) =$

$\Rightarrow = P(A \cup \underline{(B \cap A^C)} \cup \underline{(C \cap A^C \cap B^C)})$

Probability calculation: continuous example

- (x, y) such that $0 \leq x , y \leq 1$
- **Uniform** probability law: Probability = Area



➡ $P(\{(x, y) \mid x + y \leq ½\}) = \frac{1}{2} * \frac{1}{2} * \frac{1}{2} = \frac{1}{8}$

➡ $P(\{(0.5, 0.3)\}) = 0$

$P(\{(x, y) \mid x + y = ½\}) = ?$